image is essentially a standalone, executable software package comprising all the tools, libraries, settings and binaries required to run the code on (almost) any given operating environment. It helps streamline a number of tasks that would otherwise lead to considerable overheads in terms of man-hours spent on setting them up.

Containers vs virtualisation

Containers provide an isolated environment—an operating system-level virtualisation—with their own process and network space. This is far lighter than a virtual machine, which provides full virtualisation in an attempt to replicate the functionality of a physical computer. Virtual machines characteristically use a hypervisor for native execution, in order to share and manage hardware, allowing for multiple isolated environments that are sharing the same physical machine.

There are a number of trade-offs between both virtualisation and containerisation but the primary difference is as shown in Figure 3.

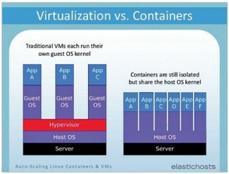

Figure 3: Virtualisation versus containers (courtesy: *ElasticHosts*)

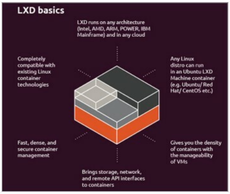

Figure 4: Orchestration of Linux containers using LXD (courtesy: *ubuntu.com*)

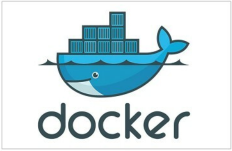

Figure 5: Docker containers (courtesy: *docker.com*)

Features of Linux containers

Containers provide a simplified pipeline for shipping a product across testing and production environments by eliminating platform-specific requirements for an application.

Lightweight and resource-friendly: They permit the running of multiple instances of applications or operating systems on a single host without much overheads for the CPU and memory, thus preserving both rack space and power.

Rapid and easy deployment: Containers make the deployment process simple enough – it is just a matter of downloading and launching multiple instances of the desired image across multiple servers. The testing phase becomes swift due to the availability of clean system images.

Process and resource isolation: The primary use-case for containers is aptly handled by Linux containers. There are multiple applications that run safely and securely on a single system—if the security of one is compromised, the others remain unaffected.

Services: As adoption rates rose, services were developed to support the idea of a container. Control groups (cgroups) is a kernel feature that allows the CPU to limit resources utilised by a process. An initialisation system (systemd) is used in setting up and managing these processes within this isolated environment. Another key feature included user name spaces, which allowed local privileges to be isolated from global ones without much modification of the environment.

Container management: LXD provides a simple, efficient means of managing multiple containers using a simple REST interface. It can handle a large number of images, including pre-made images available for a variety of Linux distributions.

Applications of Linux containers

Containers are being used by thousands of companies in order to streamline their DevOps workflows and to simplify the shipping pipeline from development to production environments. In 2015, the Open Container Initiative (OCI) was launched within the Linux Foundation "…for the express purpose of creating open industry standards around container formats and runtime."

Over the years, the Red Hat Foundation has been instrumental in promoting as well as supporting the growth and usage of containers among other open source technologies.

The inception and meteoric growth of Docker into a multi-billion dollar business is indicative of the enormous potential this technology presents for businesses worldwide.

Users of containers in production include thousands of established businesses, ranging from Google and Facebook, to Microsoft, Amazon and the smaller ones seeking to establish a foothold in the market. Say what you will … containers are here to stay! END

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